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When to use what?

- **Optimal Transport**
  - ✅ When geometry of support matters (images, shapes, distributions on a metric space).
  - ✅ When supports may not overlap (OT still finite).
  - ❌ Heavy computational/statistical cost in high dimension (except in some cases + mitigated by EOT)
- **f-Divergences** (KL, JS, χ^2, …)
  - ✅ When densities overlap and likelihood ratios are meaningful (language models, inference).
  - ✅ Strong information-theoretic foundations (variational inference, GANs).
  - ❌ KL breaks down if supports are disjoint (divergence = ∞).
- **Integral Probability Metrics** (IPMs: MMD, Energy, …)
  - ✅ When efficient, dimension-free estimation is needed (two-sample tests, kernels).
  - ✅ Flexible via choice of function class $\mathscr{F}$.
  - ❌ Geometry depends heavily on $\mathscr{F}$; may miss fine structure.

Takeaways & Preview

- OT is part of a broader family of statistical divergences, two close cousins:
  - f-divergences (likelihood ratio-based)
  - Integral Probability Metrics (witness-based)
- Choosing the right divergence = trade-off between geometry, smoothness, and tractability
- **Next time**: deeper dive into OT as an ML loss function (differentiability, mini-batching, backprop, etc)